

Smart Water Pitcher Base

[CSE123 Group 8]



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1 Introduction

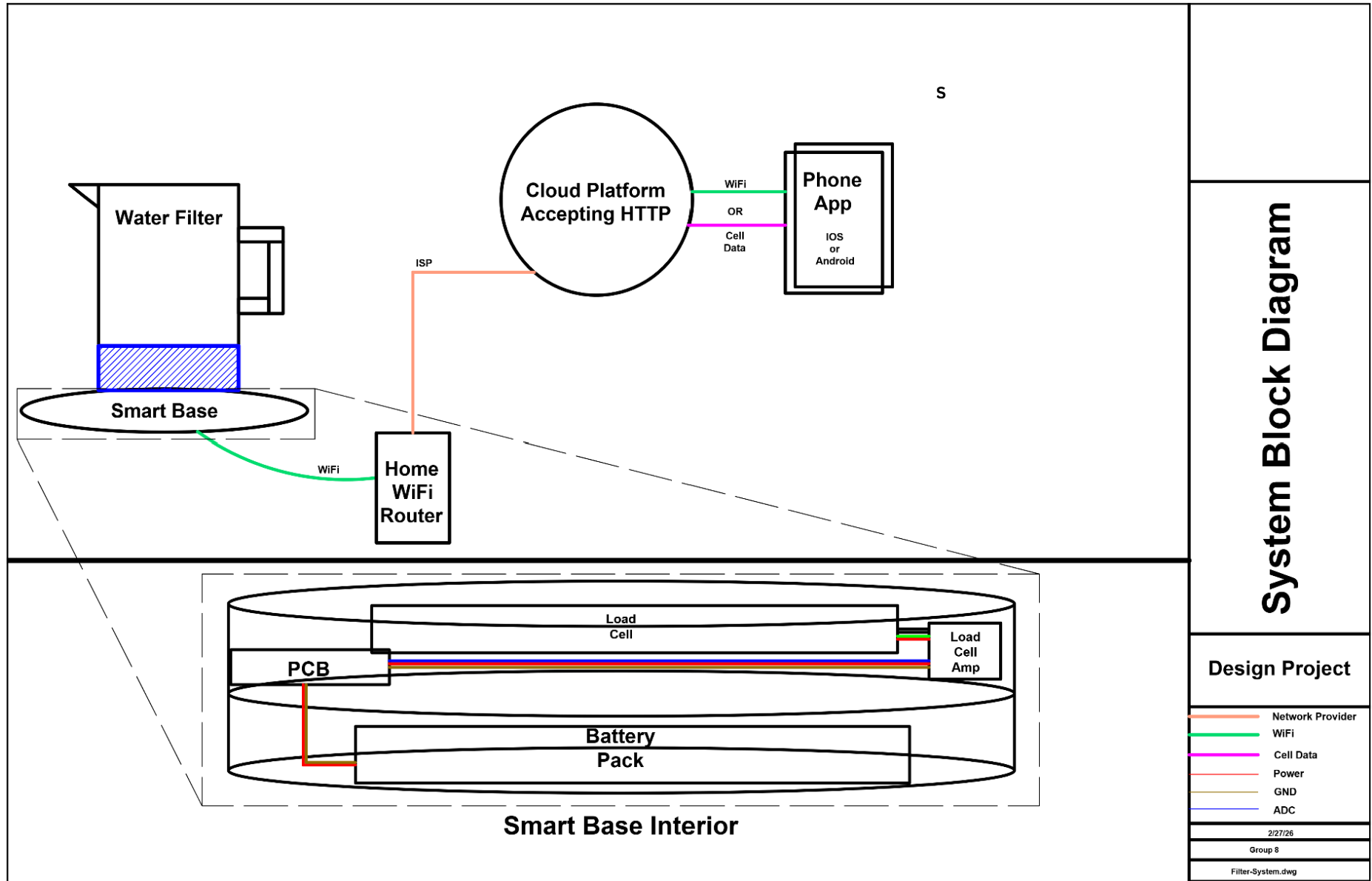
We designed a method of knowing how full your water pitcher is at any moment of the day. The design is intended to keep all users up to date with the status, as well as inform users if the level goes below a threshold.



2 System Overview

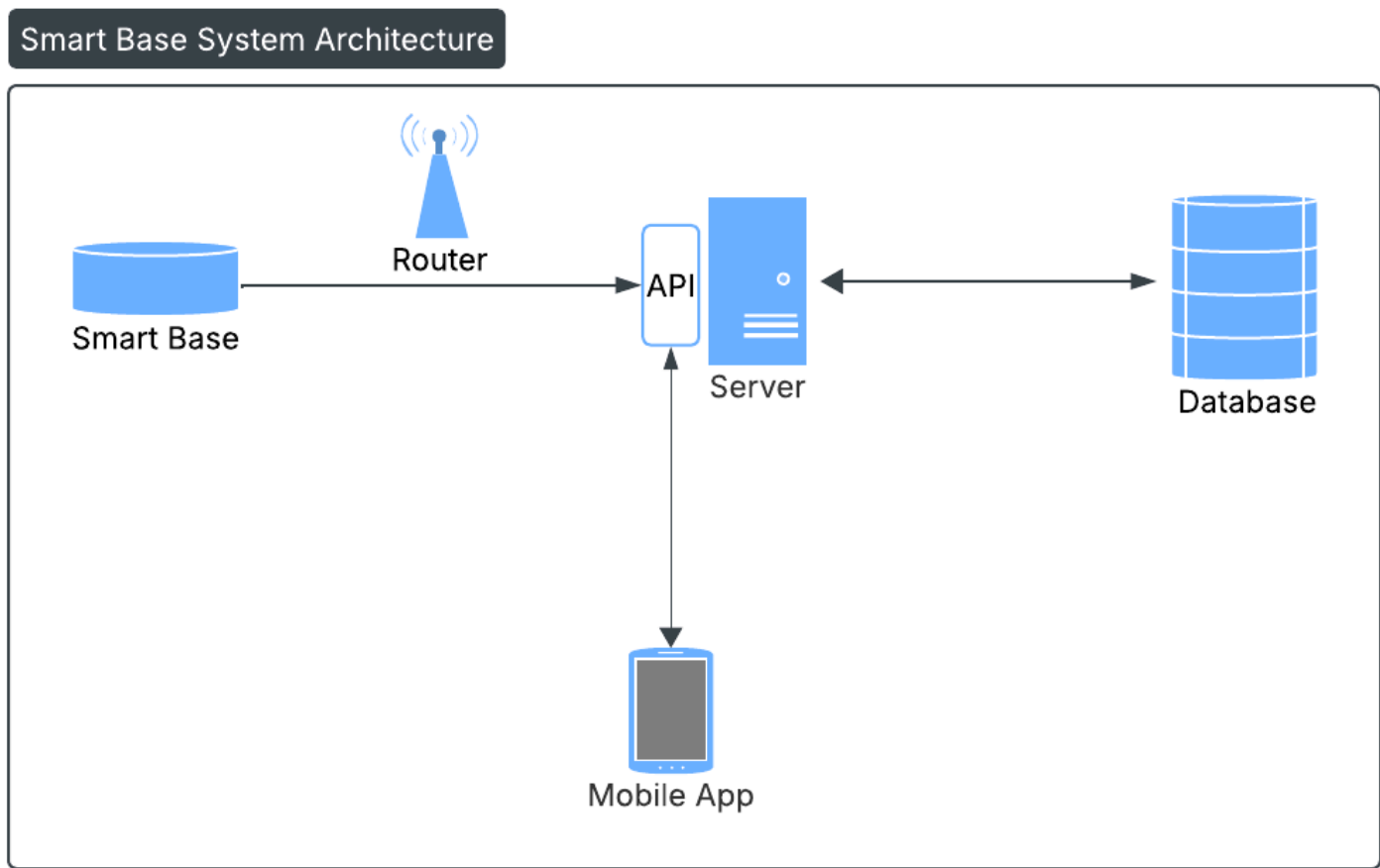
Components:

- Smart Base
- Web Server
- Database
- Mobile App

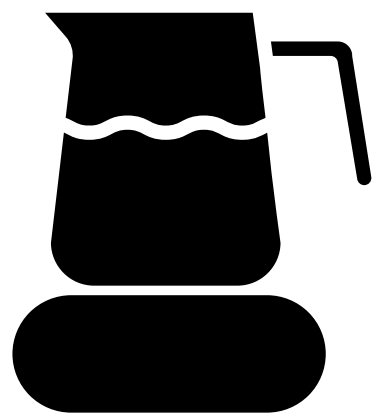


Communication

- Components communicate with in-home WiFi
- Server tracks users and devices
- Server sends alerts to mobile app to keep users up to date



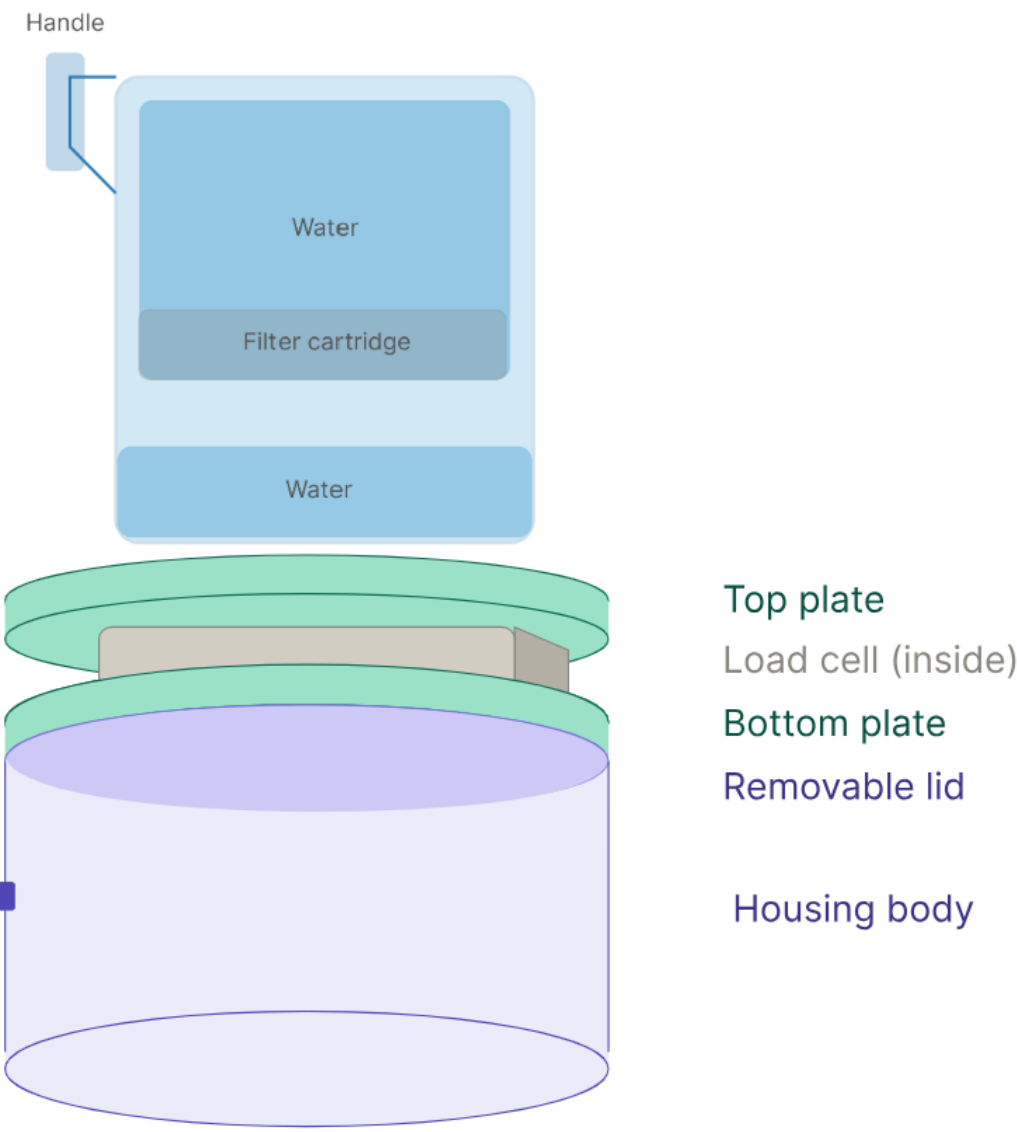
3 Smart Base



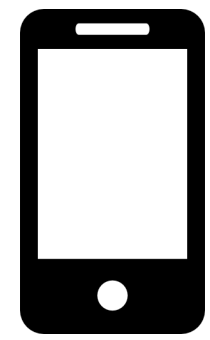
- Stays in fridge or on counter
- Water pitcher rests on top of it

Capabilities:

- Measures weight of water pitcher
- Sends weight over WiFi to server to be processed

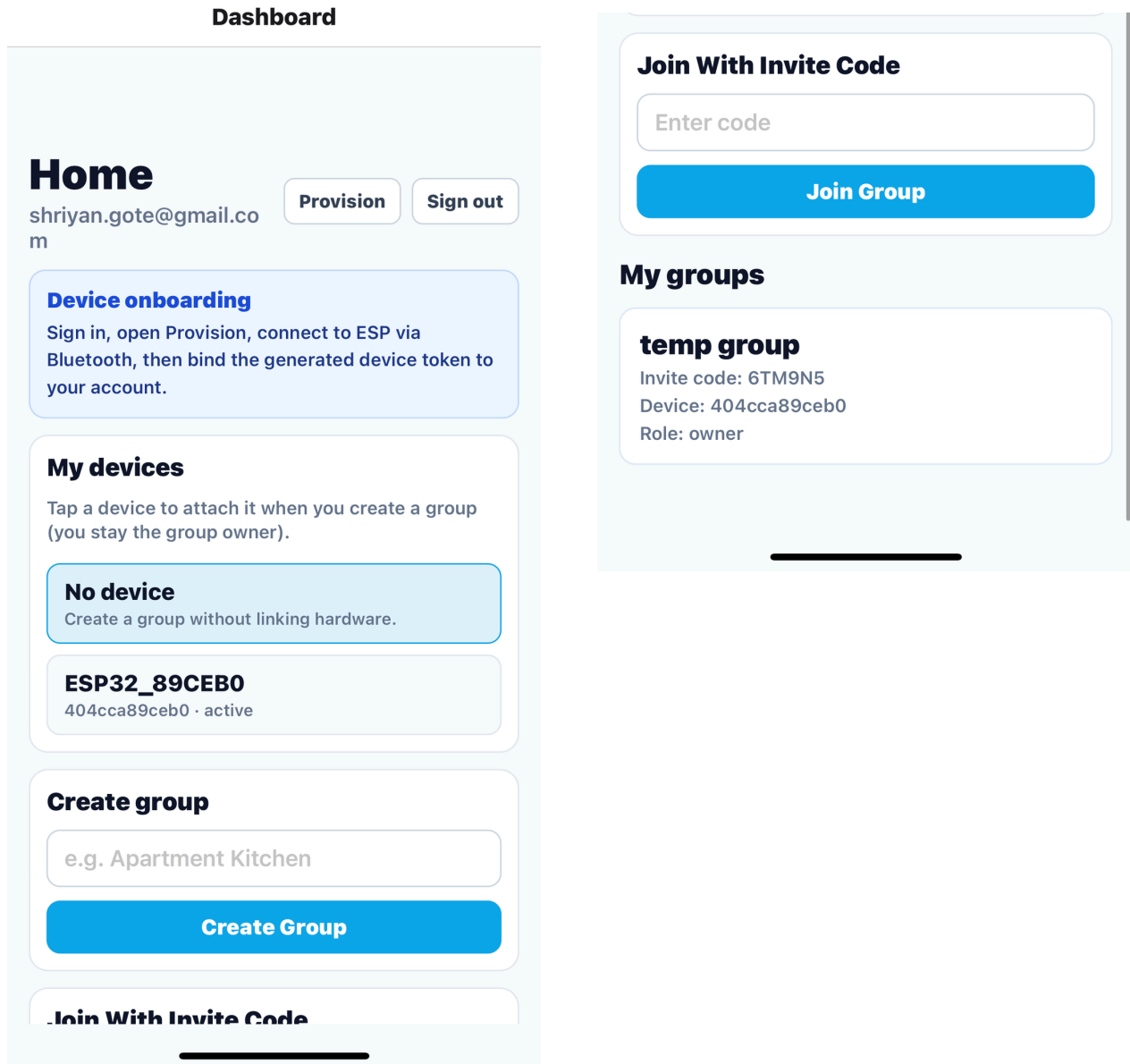


4 Mobile App



Users can:

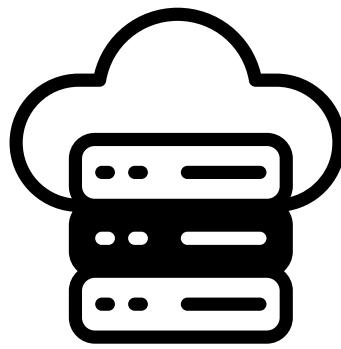
- Register multiple smart bases
- Get current water levels
- Subscribe to be notified when pitcher is close to being empty



5 Server and Database

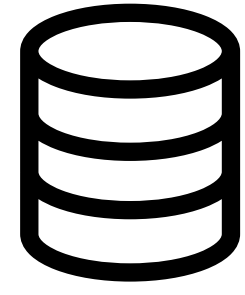
Server:

- Processes weight data, converts to fullness level
- Manages API calls



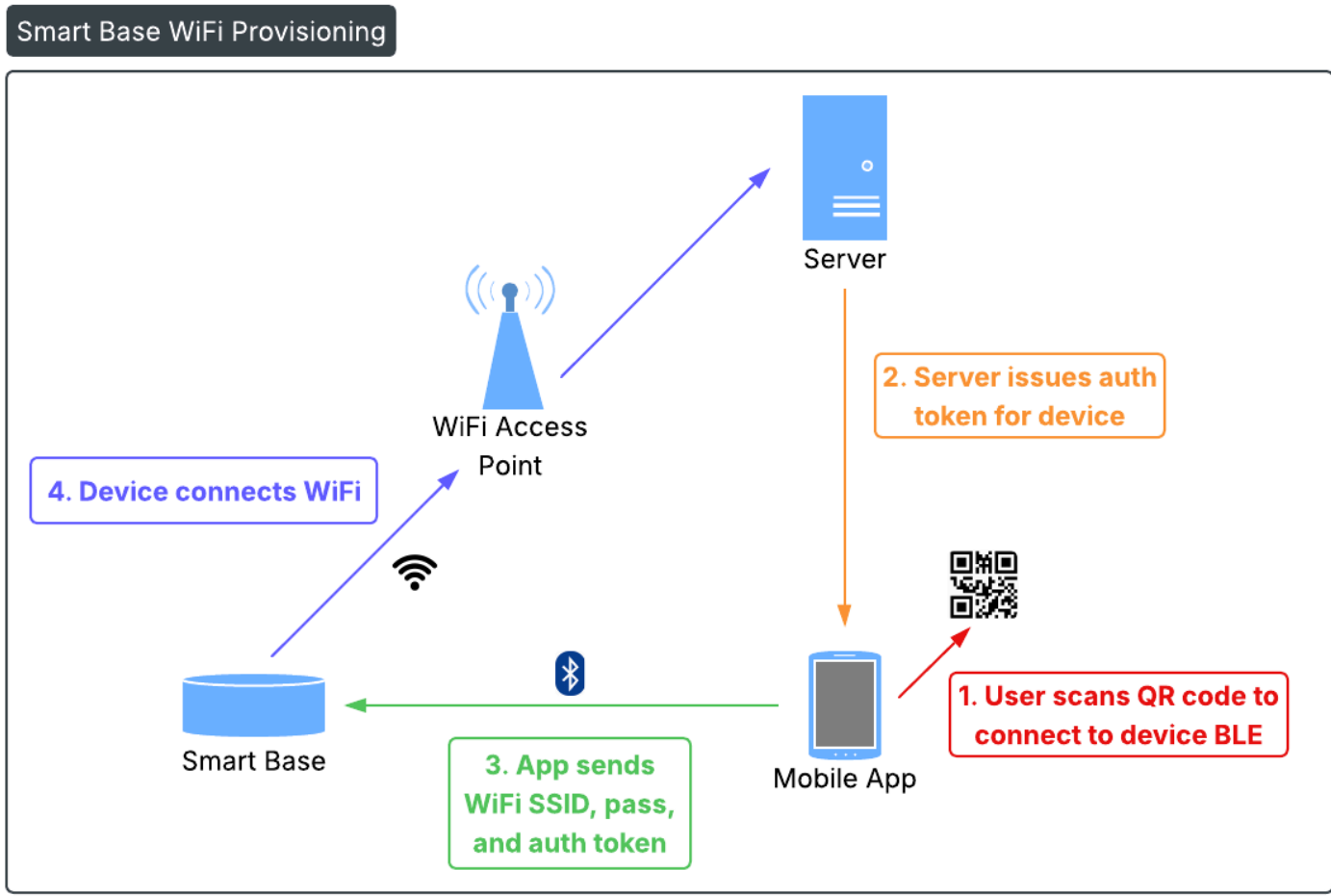
Database:

- Converts weight data to percent full
- Stores user info, smart bases, and relations



6 Usage

1. Connect smart base to WiFi network



2. Calibrate base

Calibrate empty

Calibrate full

Reset

3. Join a group on the mobile app

Join With Invite Code

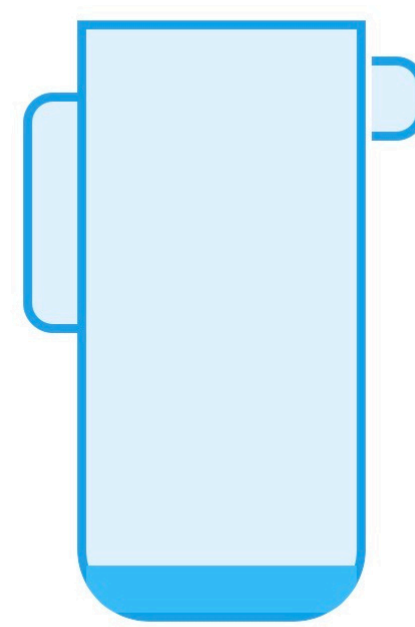
Enter code

Join Group

4. Check water level at any time

Water Filter Level

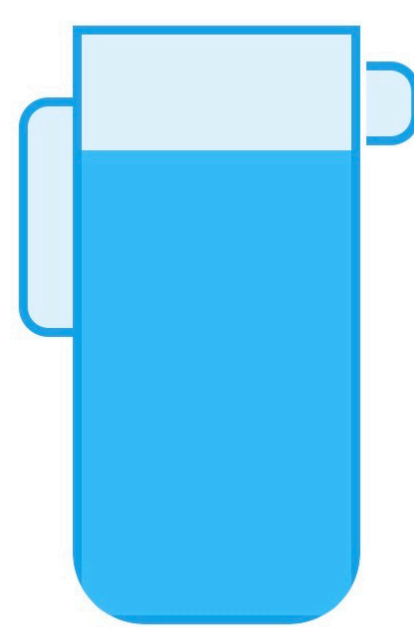
Water detected



8%

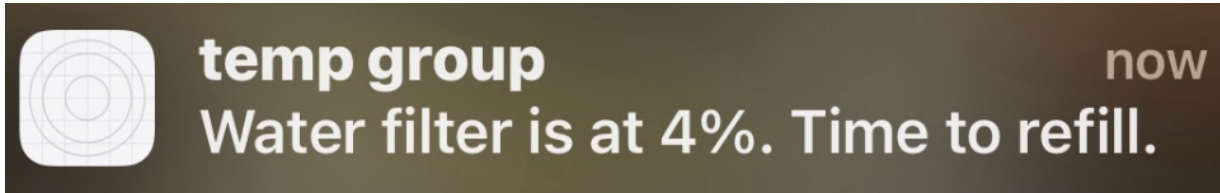
Water Filter Level

Water detected



80%

5. Receive notifications when water is low



7 Our Prototype

Smart Base:

- ESP32C3 microcontroller
- HX711 amplifier
- 3D printed structure

Mobile App & Server:

- React Native
- Vercel webserver
- Supabase database

